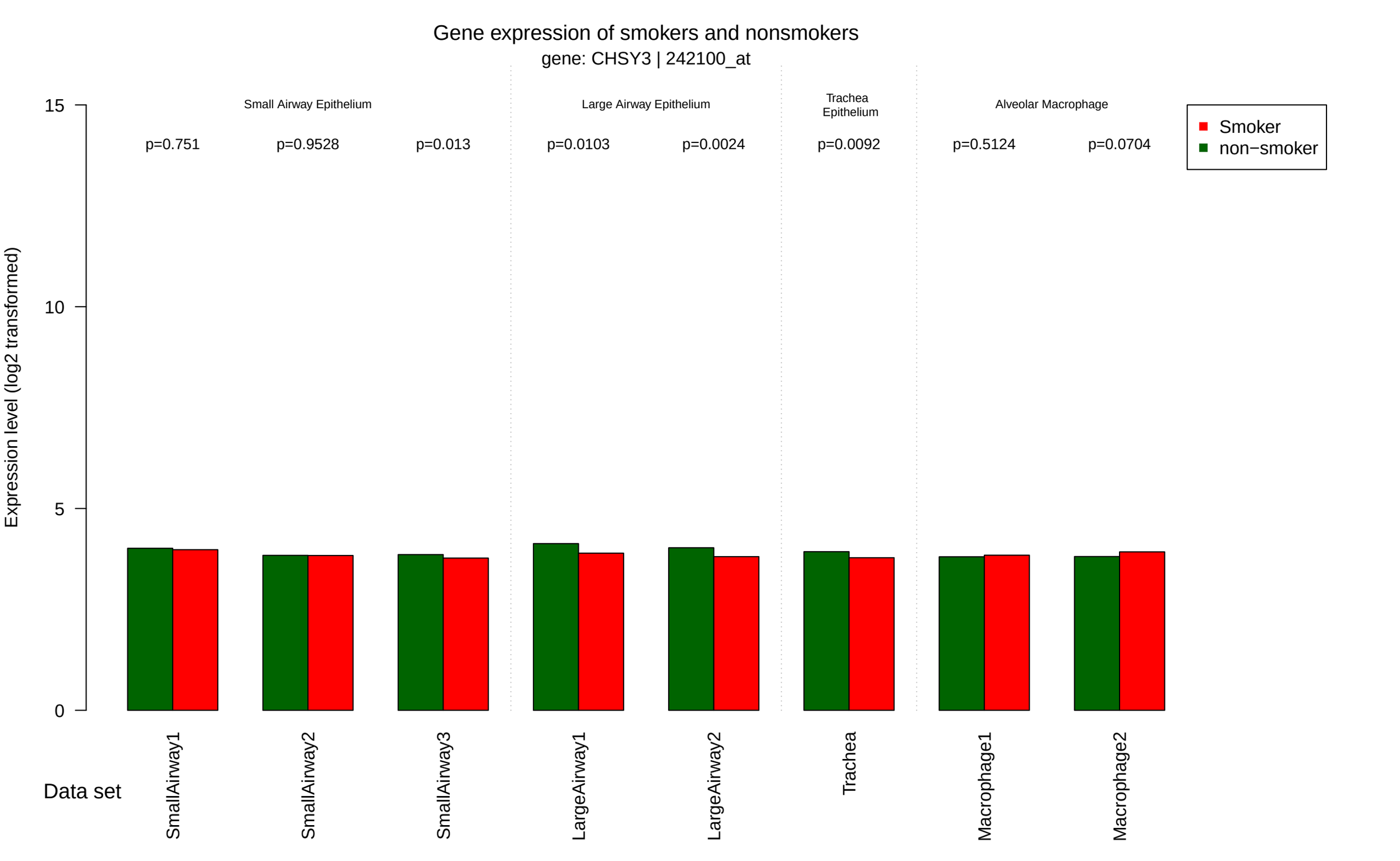
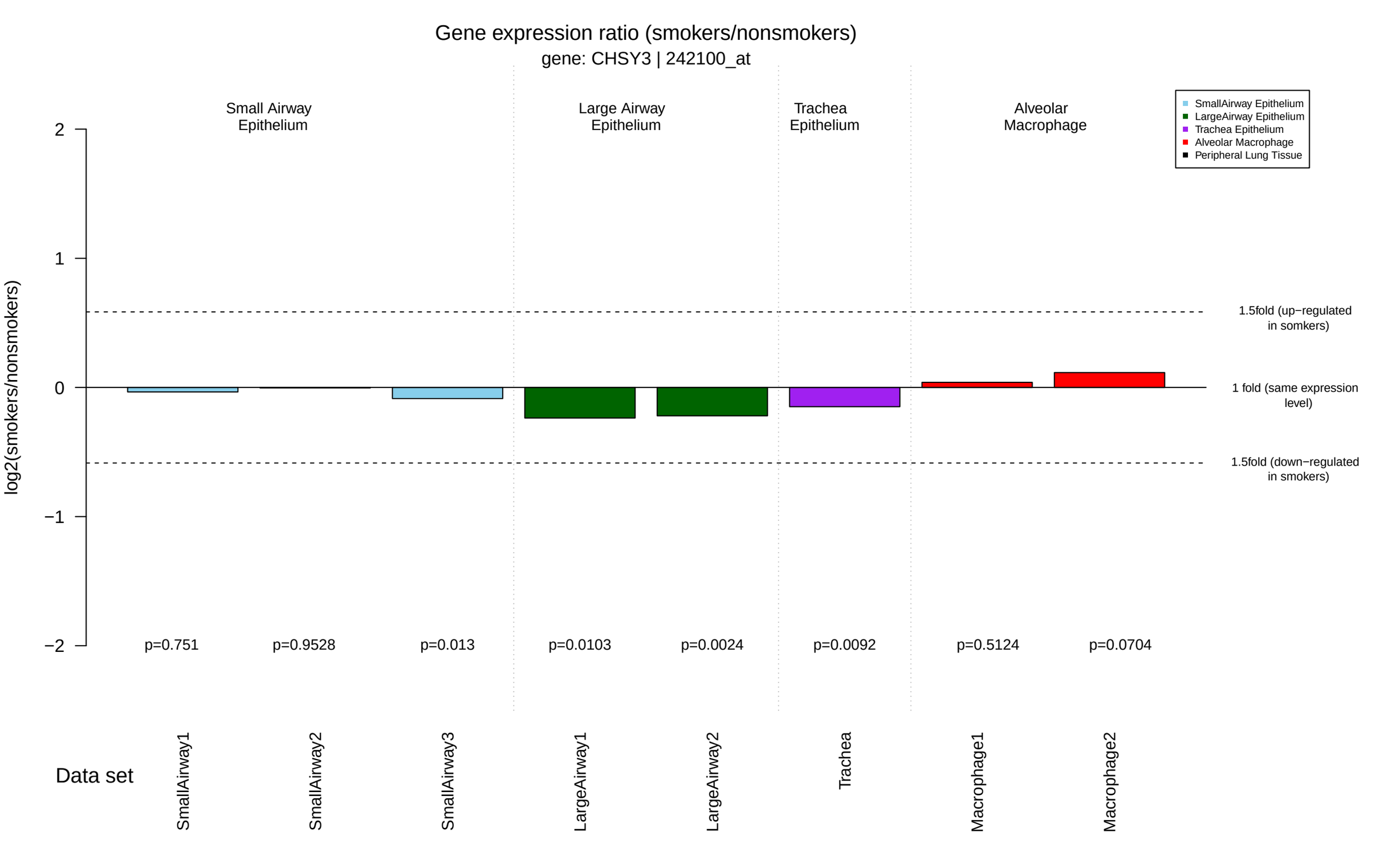
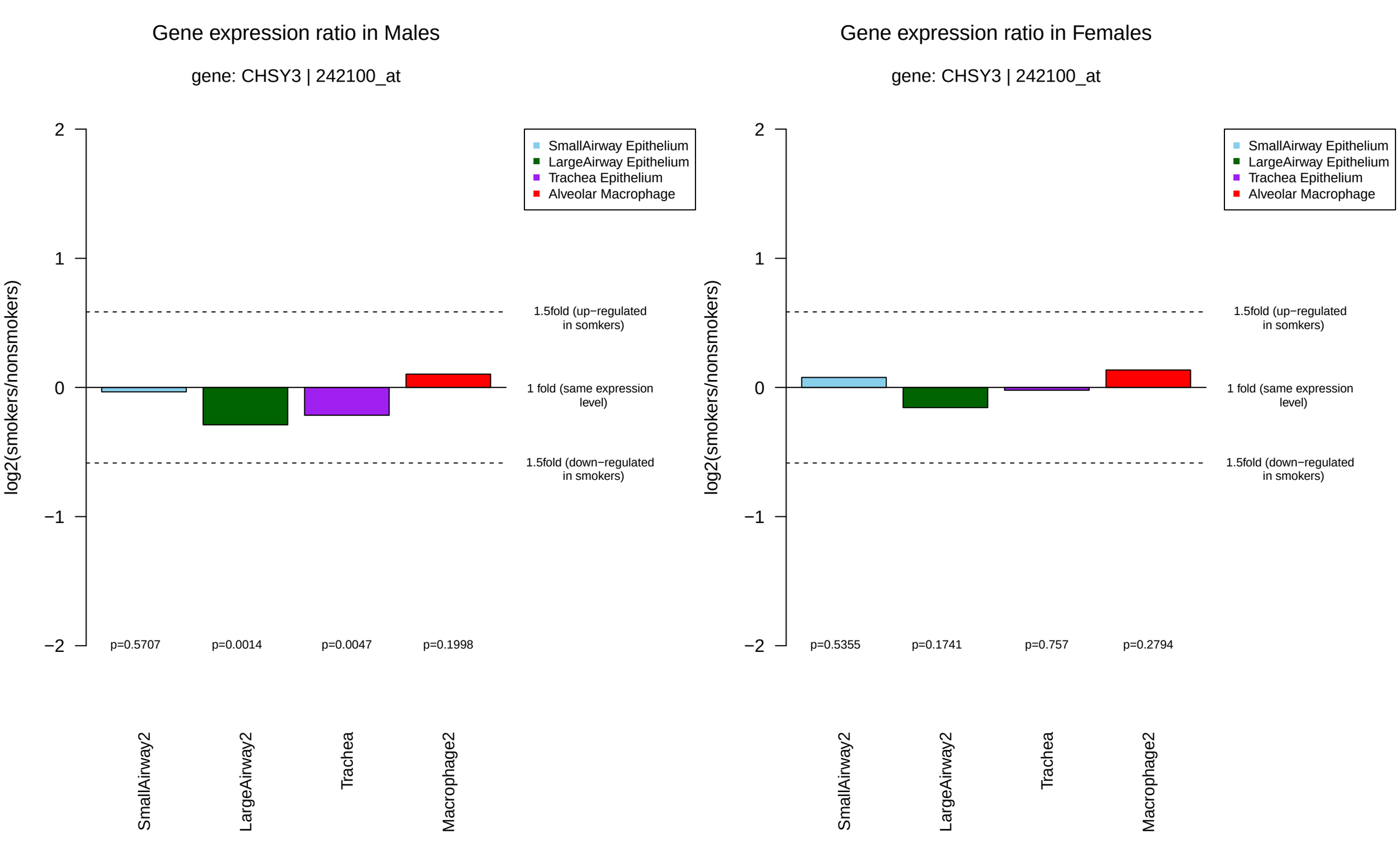
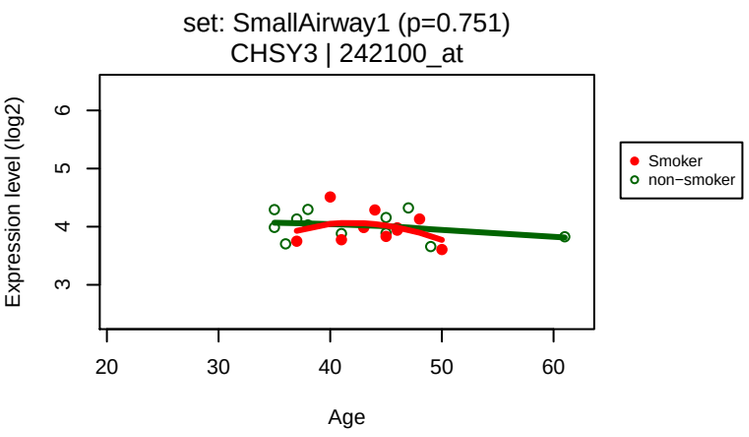
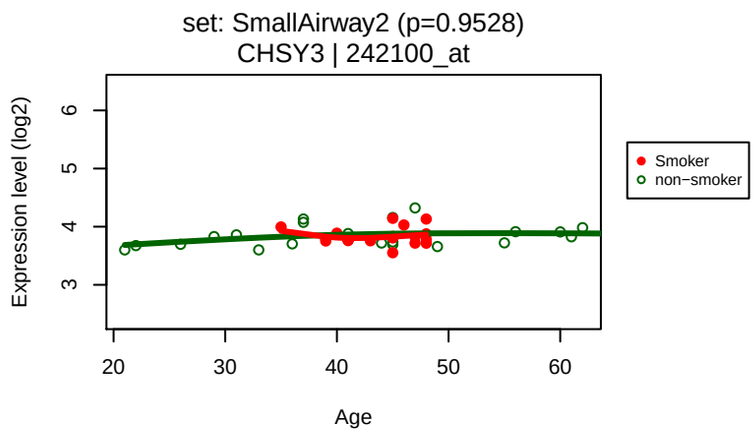
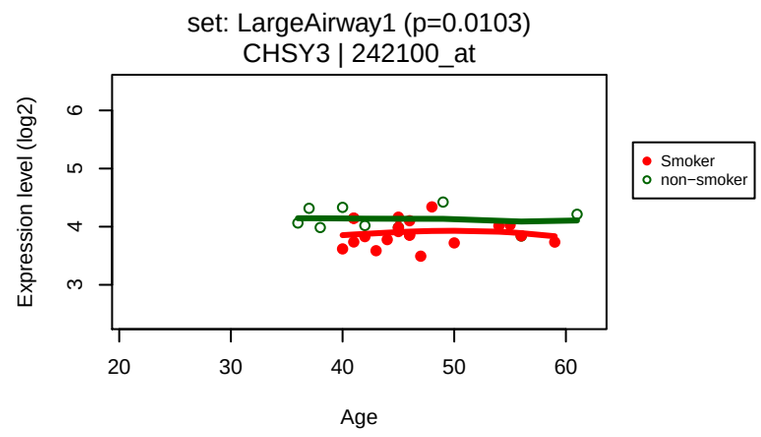

Gene expression of smokers and nonsmokers
gene: CHSY3 | 242100_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.751
p=0.9528
p=0.013
p=0.0103
p=0.0024
p=0.0092
p=0.5124
p=0.0704
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

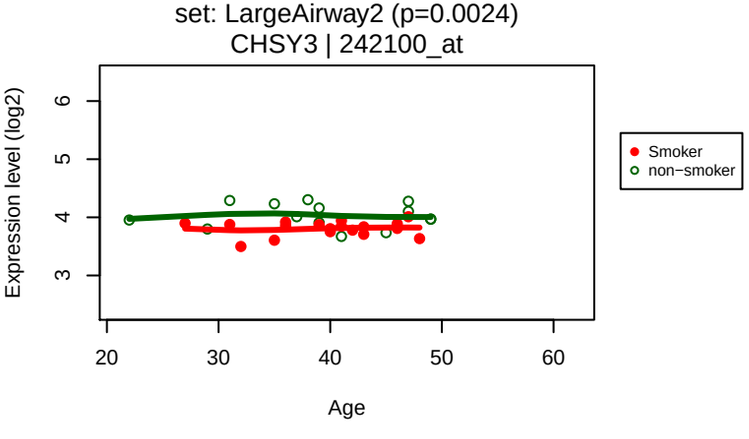

Gene expression ratio (smokers/nonsmokers)
gene: CHSY3 | 242100_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.751
p=0.9528
p=0.013
p=0.0103
p=0.0024
p=0.0092
p=0.5124
p=0.0704
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

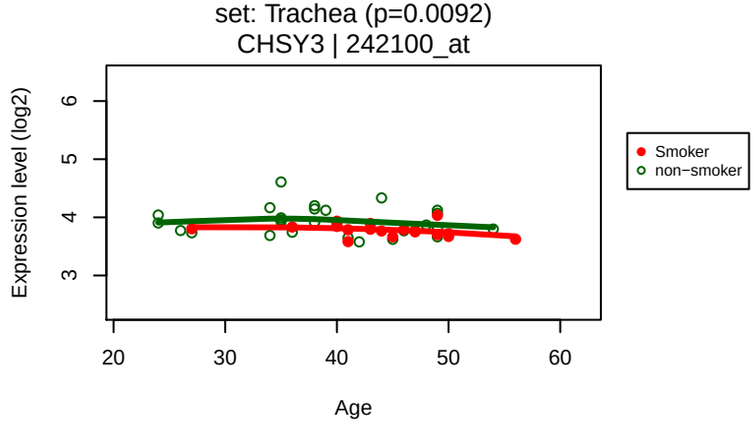

Gene expression ratio in Males
gene: CHSY3 | 242100_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5707
p=0.0014
p=0.0047
p=0.1998
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHSY3 | 242100_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5355
p=0.1741
p=0.757
p=0.2794
SmallAirway2
LargeAirway2
Trachea
Macrophage2

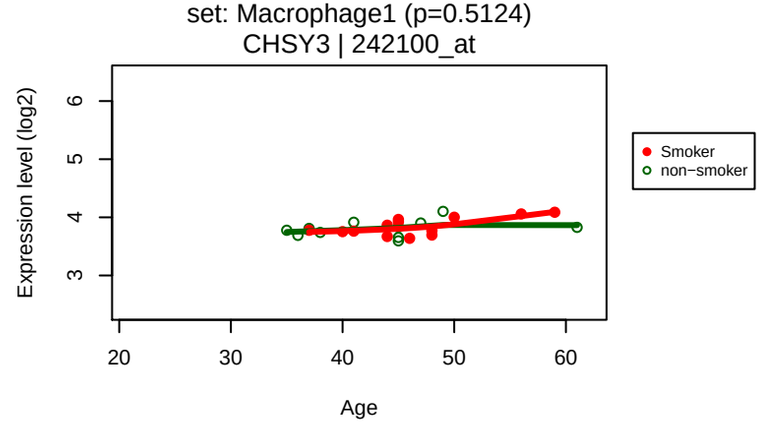

set: SmallAirway1 (p=0.751)
CHSY3 | 242100_at
Expression level (log2)
Age
Smoker
non−smoker

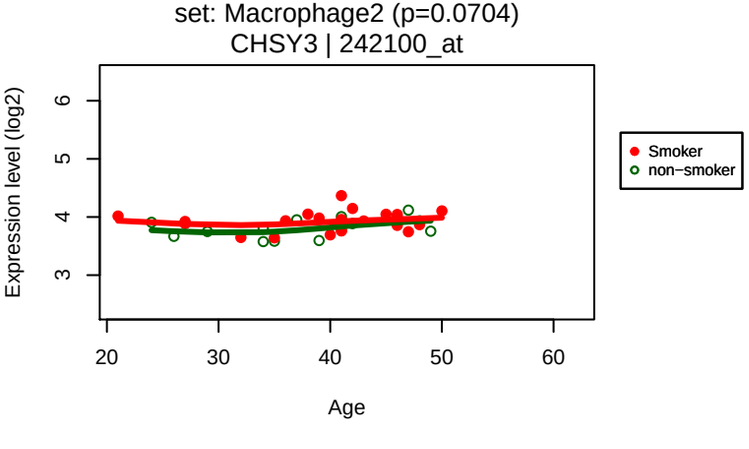

set: SmallAirway2 (p=0.9528)
CHSY3 | 242100_at
Expression level (log2)
Age
Smoker
non−smoker

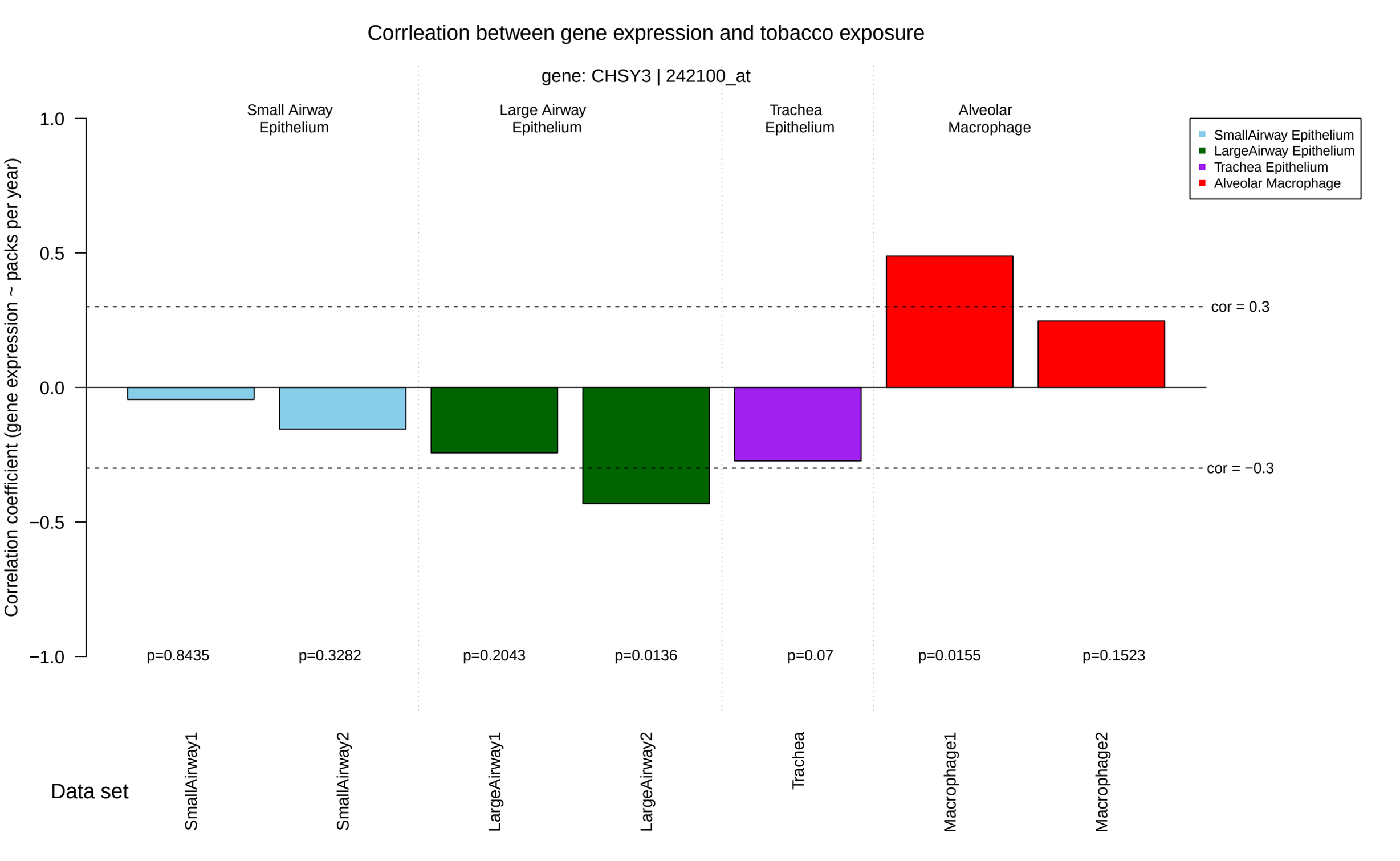

set: LargeAirway1 (p=0.0103)
CHSY3 | 242100_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0024)
CHSY3 | 242100_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0092)
CHSY3 | 242100_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5124)
CHSY3 | 242100_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0704)
CHSY3 | 242100_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHSY3 | 242100_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8435
p=0.3282
p=0.2043
p=0.0136
p=0.07
p=0.0155
p=0.1523
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2